Proceedings of the Third International Conference on Automation, Computing and Renewable Systems (ICACRS-2024)
IEEE Xplore Part Number: CFP24CB5-ART; ISBN: 979-8-3315-3242-0

2024 3rd International Conference on Automation, Computing and Renewable Systems (ICACRS) | 979-8-3315-3242-0/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICACRS62842.2024.10841656

A Review of Extracting Metadata from Scholarly Articles using Natural Language Processing (NLP)

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***Abstract*— Reasons for the need for efficient metadata maintenance and preservation in articles include the ever-growing scientific journal content. The article is designed to review a variety of different approaches for Natural Language Processing (NLP) that can be used in metadata extraction and reveals motives, aims, and undertakings associated with each of them. The study aims to find out the major milestones and common challenges in the industry through the assessment of the approaches that are in practice. To optimize the constantly increasing volume of academic information for the users, the major purpose is to provide a comprehensive overview of how academic NLP methods can be applied to automate metadata extraction. In an attempt to make such a contribution, review outlines techniques, reviews the literature, and points to possible directions for future work in the development of more scalable models.**

Keywords*— *Metadata extraction, Scholarly articles, Natural Language Processing, Information retrieval, Automated systems.

I. Introduction

The continuous increase in the number of articles being published has become a major problem in how these articles can be indexed and how relevant information can be found. The title, authors, publication date, and even abstract, known as metadata, play a crucial role in sorting and getting access to scientific articles. Since manual methods of metadata extraction include a lot of time consumption and labor intensity, there is a need to create the automated one. As for the treatment of the requirement the research has identified Natural Language Processing (NLP) which is a processing of texts based on computational techniques. The below figure 1 indicates the working flow of meta data extraction [1].

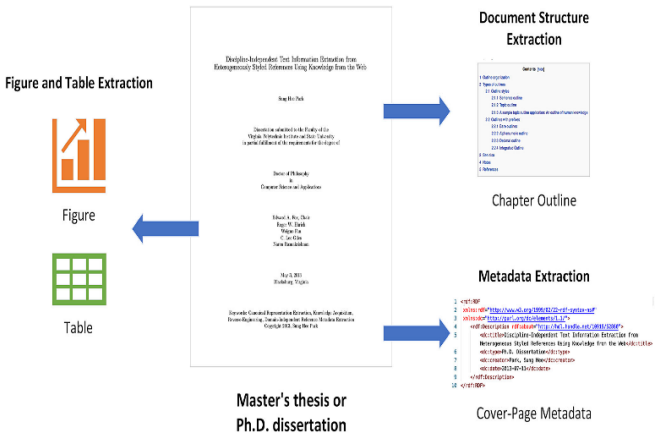

Fig. 1. Example of Meta data Extraction[1]

In the last several years, a great deal of research has been dedicated to the development of NLP algorithms which could help with metadata extraction from academic articles. Due to these efforts, there is going to be better efficiency and accuracy in Information Retrieval systems hence will enhance the discoverability of academics content. For example, Ingram et al. (2024) demonstrated the applicability of the concept of machine learning models in the course of building datasets toward facilitating the extraction of information and parsing of structures from electronic theses and dissertations [3]. Another study was made by Turgunbaev in 2024 on discovering the metadata of the academic publications with the help of the machine learning algorithms. According to the study, it was revealed that it is possible to reduce the amount of intervention and error that is required [3].

That is why despite the progress that has been achieved there are still some difficulties that have to be solved in order to achieve very high precision and scalability of metadata extraction. The difficulties of data dependence, computational costs, the requirement for constant update and the need for human intervention have not been eliminated from the system. Assuming such tasks requires possessing a broad understanding of the strategies that are already being implemented and the constrains they come with. The reason why, specific review article aims at providing a comprehensive analysis on the several approaches to metadata extraction within the context of NLP. Further, it will propose directions for future research for the improvement of the performance and reliability of such systems as well as the differences between the approaches used by different scholars.

II. Related Works

Ingram et al. [1] describe approaches to building of the datasets to support the extraction and modeling of the information from the thesis and dissertations. The authors focus on the difficulties of text processing, by converting academic publications, which are rather unstructured, into a format that will allow using more complex machine learning algorithms. They outline an approach that can be used for data mining that also addresses the enhancement of metadata in the academic repositories. Their technique underlines an importance of correct information search to improve its access and to preserve scholarly information for future use.

Turgunbaev [2] offers a model with the ML algorithms to extract metadata of an academic publication on its own. To illustrate the, the author discusses how various machine learning techniques can be used to raise awareness in increasingly structured academic texts as to what should be marked, for example, as the title, author, or other publication

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data. Based on the empirical evidence, the research proves that these models can increase the efficacy of the metadata extraction process and thereby develop scholarly databases. The paper also responds to the applicability of these techniques in the different domain of the literature.

In [3], Rizqi et al. target at enhancing the plugin that aims at extracting metadata for the Open Journal System (OJS). Thus, it is an attempt to offer an automated mechanism for extracting metadata while minimising the involvement of the journal administrators within the context of OJS. The paper also elicits the technical architecture of the plugin and how it may be deployed to real productive use to enhance crucial metadata harvesting including the title of the article, the name of the author and keywords to the search. The following are the ways their system is beneficial to the management of academic journals: Their metadata about published articles are made more standardized and accurate.

LLMs are discussed by Dagdelen et al. [4] regarding their application to improving the extraction of structured information from the scientific articles. The authors pay particular attention to bandwidth of these models in terms of their ability to find, understand and abstract information such as the experimental arrangements and outcomes as well as methodologies, from scholarly publications. They draw attention to the possibility of using full LLMs in enhancing the performance of information retrieval and the time taken in the process especially for the scientific world where one is required to go through huge volumes of documents. The study shows that use of LLMs has potential in increasing its efficiency in the scientific field for automating and enhancing the data extraction process.

There are so many approaches and methods that can be employed in the resolution of the above problem and some of them include: Iwashokun and Ade-Ibijola [5] developed a method to parse research documents into XML using formal grammars. Such work stresses the need to use standards such as XML when preparing data with the aim of enhancing their findability and retrievability to readers. Looking at the case of using the formal grammars, the authors manage to systematically parse complicated and unstructured documents into structured forms. The presented approach offers the base for increasing the metadata production rate and enhancing the organization and search procedures of research documents in digital libraries.

Butcher et al. [6] look into improving the interaction between human and machine to improve the accuracy of information retrieval from texts. In view of the, the authors propose a enriched model that combines both human and artificial intelligence in achieving high accuracy data extraction especially where the text is sensitive, difficult or contains overlapping values. Their studies are compatible with the hypothesis that the integration of a machine and a human into the common workflow to complete the latter's work, where the machine processes all regular cases, and the human solves unusual extraction problems, is effective for improving results. The applicability of the methodology goes a long way in enhancing data handling in all disciplines that need accuracy in handling and analyzing texts.

Liu and Li [7] propose an automatic method for mining existing know-ledge from text mines particularly, cutting down the manpower effort. The whole framework involves several methods of machine learning for content curation, which includes content sources, research findings, methodologies that are used in the research papers, as well as citations. The the authors stressed the fact that their approach can easily be extended to large corpora of academic text. They stress that with the help of the framework the speed of data extraction in academic and research contexts can be increased many times.

Another remarkable study by Saier et al. [8] propose a new system aimed at the extraction of hyperparameters from scientific papers that is so important for reproducible machine learning. The system operates with the texts in the field of science and applies the NLP methods to distinguish the hyperparameters from the texts' bodies. The authors describe difficulties of text comprehension and relevance of correct identification of features that may influence the replication of experiments. The system proves to have positive impacts on the increase of the efficiency and accuracy of the extraction of hyperparameters.

Suryani et al. [9] has presented a data capture architecture that will enable information to be extracted from scientific publications. The authors provide a systematic approach to harvest and index data from scientific paper which will aid in offering better and extensive data from retrieval. Comparing with conventional methods on big data analysis, it enhances the efficiency of extracting crucial information from academic texts by introducing machine learning models alongside existing data acquisition approaches. The framework is quite helpful with broad data acquisition campaigns in any field of scientific inquiry requiring the compilation of specific details.

Syntactic features enriched transformers implemented by Bölücü et al. [10] focus on the effects of extracting quantities present in the scientific body of texts. The authors give their insight into how they work towards enhancing transformers; a type of machine learning model to understand and extract numerical information from difficult technical articles that include syntactic facets. From the suggested equations, they demonstrate that syntax-derived features bring about a massive improvement in the model ability to quantify accurately, which is Significantly important in engineering and science fields that require exact value extraction.

Another study completed by Ikoma and Matsubara [11] discusses the application of language models for determining the roles of citations in articles with reference to how citation is utilized in scientific language. To the end, the authors build a taxonomic system of citation classification that defines what type of function citation fulfills in the context of text. In the way, the work illustrates that automated systems can consider citation behaviour so enhancing the knowledge of scholarly communication. One must add that their approach opens up a new perspective in the study of the processes of academic referencing.

Tikhobaeva and Bruches [12] concentrate on relation extraction from Russian scientific texts only and specially discuss problems of a small training set. They propose a system for dealing with the lack of rich annotated datasets, using such techniques as transfer learning. The authors prove that using their approach leads to a high improvement of extracting relations between entities, like authors, institutions, topics, even with a small amount of data. To the best of the author's knowledge, the present work joins a line of ongoing efforts directed at enhancing the efficiency of information

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extraction from scientific literature in languages other than English.

In the technique Marshalova et al. [13] discuss extraction of components from scientific literature; particularly, they emphasize identification of key components such as study objectives, methods and results. These aspects are categorized and extracted in an automatic manner from a collection of unstructured academic text using natural language processing that the authors use in their papers. The system their help is to help researchers in decreasing the time they spend in pondering over papers manually and in increasing the likelihood of identifying essential elements. The findings suggest that automation is favourable toward enhancing the processes of academic research in particular.

Panayi et al. [14] assess a machine learning tool devised for partially automating the task of data collection for SLRs. The application is intended to help filter large amounts of scholarly beneficiation by autonomously identifying primary data components, including study characteristics and results. The authors reflect the usefulness of the tool in terms of the reduction of the load on the researchers' hands and the increase in the accuracy of the extracted data. According to their results, it can be stated that the use of semi-automated tools may improve the effectiveness of systematic reviews in various fields of study.

Foppiano et al. [15] describe a semi-automatic system which allows the creation of high quality structured data from scientific documents. Their system incorporates machine learning algorithms and human annotations and reviews to achieve highest level of precision in data extraction especially these fields where precise data is needed for meta-analysis or systematic reviews. Still, it also allows keeping the data clear from human bias to ensure its quality and, at the same time, demonstrate how it improves the efficiency and accuracy of structured data retrieval from complex scientific articles.

III. Differents Methodology

A. NER stands for "Named Entity Recognition"

The Named Entity Recognition (NER) technique, is a known NLP technique which is used in order to detect items in text and categorize them for instance into names, dates or organizations. NER can be used in recognizing some of the components automatically from the scientific papers when applied in the process of extracting metadata. Such components may be an author, the date of publication, or the institution that was used in the work. For example, the case that examined by Ingram and colleagues (2024), they demonstrated that NER can be used to effectively extract information from formatted electronic theses and dissertations [2]. NER systems can reach a high percent of success in locating as well as categorizing the proper items inside the academic texts. The is done with the help of training models with the help of annotated datasets.

B. Approaches that are based on machine learning

In the process of metadata extraction and the use of machine learning algorithms particularly the supervised learning techniques, they have been applied. In these methods, models work on the dataset that has been pre-labeled so as to look for specific features on the data that is of most significance. Turgunbaev (2024) revealed that several methods based on machine learning like support vector machines, and random forest were considered in order to identify metadata of academic publications automatically [3]. The use of these methods enables the development of models that are extendable to other data sets, something that enhances scalability and efficiency of some of the processes of information extraction.

C. Structures Based on Rules

Rule-based systems are used for the process of identifying and extracting of the metadata from the text by using the ready sets of rules. These approaches are particularly useful where there is a clear structural pattern within the text, and the does not change. Iwashokun and Ade-Ibijola in their research have employed formal grammars to transform research documents to XML (Iwashokun and Ade-Ibijola, 2024). The was done largely with an aim of showcasing the efficiency of the rule-based techniques when it comes to structured IE processes [6]. Rule-based systems are couped of a high level of precision; on the other hand, the systems can be rigid sometimes, and it might take considerable time and effort to integrate the current data into the systems and to continue updating them.

D. Different Methods of Deep Learning

Maximising the use of natural language processing, deep learning models including neural networks along with transformer models have been shown to be a lot effective in the field of information extraction. I pointed out that these models can identify even delicate dependencies and relations within the text and, therefore, produce more accurate and reliable metadata. Some of the recent studies on the employ of big language models include the work done by Dagdelen et al. (2024) which focuses on how BERT and GPT models were used to extract structured information from the scientific text while making high levels of precision and scalability achievable [5]. The has brought close to nil chances of manual interjection and frequent employment of upgrades in cases of deep learning models.

E. Using a Hybrid Approach

In the way, different strategies are utilized simultaneously so as to enhance the advantages of one approach and practically remove the flaws of the other. For example, Butcher et al. (2024) proposed a way of using machine learning model while at the same time, affording human intervention to enhance precision [7]. The makes it an ideal approach for enganging human and machine skills in a way that optimally matches human precision in the selection of information to extract. Due to the capability to maintain the balance between automation and accuracy these methods are suitable for the extraction of such a highly complex metadata.

F. Frameworks that are Automated

It is noteworthy that currently there are automated frameworks, such as AutoIE, described by Liu and Li (2024), that are aimed at providing end-to-end solutions in metadata extraction. To make the whole extraction process fully automatic, these frameworks are designed with NLP techniques such as NER, ML, DL, and other complementing techniques [8]. The most promising aspect of such of automated frameworks that makes the scalability factor shift higher is that it is able to handle large chunk of text hence reducing the impact of manual intervention.

G. Transforms that are enriched with syntax

To improve metadata extraction accuracy, syntax-enriched transformers incorporate syntactic information into

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transformers' models. Indeed, in their study, Bolucu et al. (2024) investigated how syntax-enriched transformers could influence the process of quantity extraction from scientific texts. These GHk identified in these models confirmed their effectiveness in identifying network structures inherent within text [11]. It is because including syntactic data brings these models to a better data accuracy in terms of the identification and indexing of important information characteristics.

H. Optimisation of Hyperparameter Setting

In order to attain improved results from a machine learning model, hyperparameters are tuned for optimization The optimization of hyperparameters is important for attaining the best level of performance from a machine learning model. Still, HyperPIE is a framework that was developed by Saier, et al in the year 2024 for the sole objective of learning hyperparameters from scholarly articles. The framework also focuses on the proportioning of the model parameters, so as to achieve high accuracy [9]. It has been found that the fine tuning of hyperparameters significantly enhances the efficiency of these information extraction models and the models become less sensitive and more comprehensible.

I. Frameworks for the Acquisition of Data

In the context, the so-called data acquisition frameworks are designed especially to provide the optimal collecting and processing of data for the purpose of the metadata extraction. To enhance the accuracy and efficiency of metadata extraction procedures, Suryani et al. (2023) developed the Data Acquisition Framework (DAF) which has been that contain several techniques of natural language processing (NLP) [10]. The application of such frameworks provides a regular way of data harvesting and pre-processing thus ensuring that only high quality data are fed into the metadata extraction algorithms.

J. An Extraction of Aspects

The general goal of the aspect extraction technologies, which can be also named as contextual or template-based technologies, is to identify and extract specific aspects or attributes from the text. Marshalova et al. (2023) pointed that a methodology was developed for using text mining to identify aspects from scientific documents. The approach showed the competence of natural language processing techniques in searching and retrieving portions from large amount of content [8]. A specific way of refining the metadata extraction is aspect extraction because it is able to give deeper insights into certain aspects of the scholarly papers.

K. Partially Automated Equipment

It has also been noted that high quality metadata extraction can be done by Semi – automatic approaches which combine both the Automated as well as Manual extraction. Panayi et al. (2023) performed an assessment of a proof of concept of machine learning function to support the systematic method of data extraction for SLRs. As shown by the outcomes of the evaluation, there are benefits of combining between automated and manual processes [15]. The means that given technologies are well suited to the identification of jobs that require the extraction of very complex forms of metadata since they are capable of finding a balance between speed and accuracy.

L. Sites for the Extraction of Data Staging Areas

Staging areas for data extraction are suitable as they create an environment through which high quality data can be processed. The high need for quality data for enhanced metadata extraction was noted by Foppiano et al. (2023) who developed a semi-automatic staging area for the extraction of the structured data from the scientific writing [16]. It becomes possible to note that staging zones, on the whole, may contribute to the enhancement of overall fidelity of the metadata extraction models since it provides the assurance of the quality of the incoming data and the appropriateness of preprocessing.

IV. COMPARATIVE STUDY

TABLE I. COMPARATIVE ANALYSIS OF EXISTING METHODS

Algorithms	**Advantage**	**Limitation/Future Work**
NER [2]	High accuracy in identifying entities	Requires large, annotated datasets
Machine Learning [3]	Scalable and efficient	Data dependency and computational costs
Rule-Based Systems [6]	High precision	Lack flexibility and require extensive manual effort
Deep Learning [5]	Captures complex patterns	High computational costs and need for large datasets
Hybrid Approaches [7]	Balance between automation and precision	Requires human oversight
Automated Frameworks [8]	Efficient and scalable	Initial setup complexity
Syntax-Enriched Transformers [11]	High precision with syntactic information	Complexity in model training
Hyperparameter Optimization [9]	Enhances model performance	Computationally intensive
Data Acquisition Frameworks [10]	Structured approach to data collection	Initial setup complexity

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Aspect Extraction [14]	Detailed insights into specific elements	Requires sophisticated NLP techniques
Semi-Automated Tools [15]	Balance between efficiency and precision	Continuous updates needed
Staging Areas [16]	Ensures high-quality input data	Setup and maintenance efforts

CONCLUSION AND FUTURE SCOPE

The review also notes that the extraction of information from the scientific publications is well solved, whereas some challenges, such as data dependency, high computational costs, and the need for manual interference, still exist. Automated metadata extraction has drawn attention in natural language processing (NLP) setting but the task is difficult to scale up to large volumes number without sacrificing quality. These challenges can only be met by proper algorithm that enjoys reliable data and better framework. Recent studies should focus on the highly reproducible, affordable solutions that do not demand intervention from people. Furthermore, the integration of fully automated and hybrid techniques with lots of human interventions may extend performance by combining the elements of both, thereby aspiring to substitute metadata extraction and identification results efficiently.

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